

Separating mixtures

MIXTURES ARE MADE UP OF SEPARATE SUBSTANCES.

The constituents of mixtures are not chemically joined. Since they remain distinct substances, they can be separated using only physical means. The precise method depends on the type of mixture.

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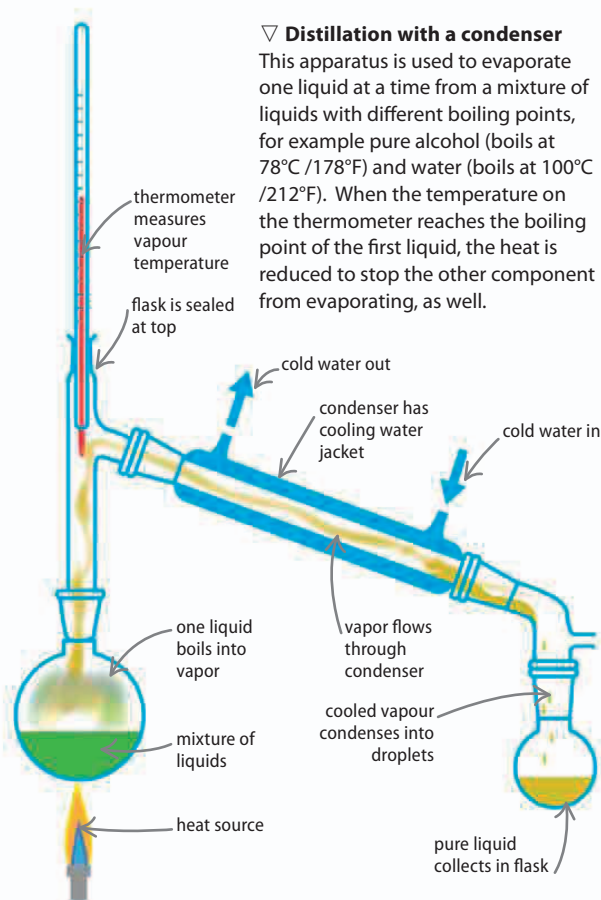
152–153 ▶

Crude oil distillation

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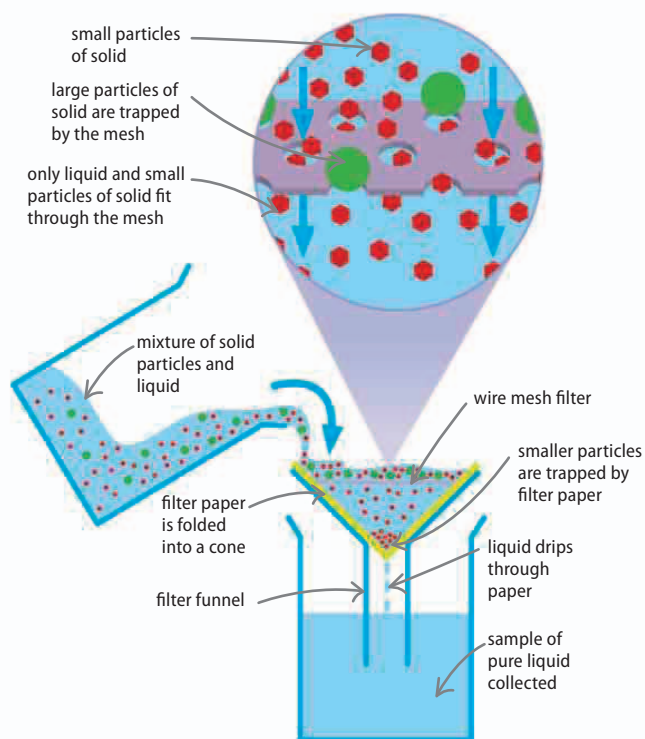
Liquid mixtures

Dissolved solids can be separated by evaporating away the liquid solvent, leaving crystals of solid behind. This is how salt is separated from seawater. Collecting a pure sample of the solvent is more complicated. The vapor passes through a condenser, where it is cooled back into a liquid. A condenser is also used in distillation, which separates a mixture of two or more liquids.



Filtration

Silt in river water is an example of an uneven mixture—large, heavy solids are mixed into a continuous medium of much smaller particles. This kind of mixture can be separated using filters. A filter is a material that allows the smaller particles through but blocks the progress of the larger ones. Most laboratory filters are made from paper, but wire meshes can be used too.



△ Double filter

The experiment above uses two different filters to separate out two different-sized constituents. Water and small particles pass through the first filter, made of wire mesh, but larger particles are trapped. The smaller particles are trapped in the second filter, made of paper, leaving just the pure liquid to drip into the beaker.